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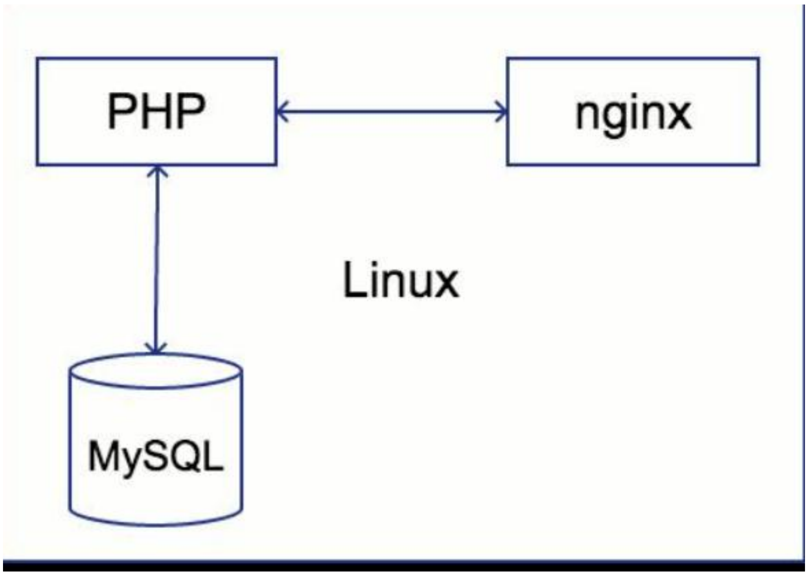
Project : 14

NexHost : Cloud-based LEMP dashboard with RDS

Architecture Diagram



Project Overview

NexHost is a cloud-hosted web application that demonstrates deployment of a **LEMP stack (Linux, Nginx, MySQL, PHP)** on AWS.
The project integrates a **frontend dashboard**, backend PHP logic, and a managed database using **Amazon RDS**, providing a complete real-world hosting environment.

🧰 **Technologies & Services Used**

- **Amazon EC2** → Linux server hosting
- **Amazon RDS** → Managed MySQL database
- **Nginx** → High-performance web server
- **PHP (FPM)** → Backend processing
- **MySQL** → Data storage **Implementation Steps**
- 1️⃣ **Launch EC2 Instance**
- Create Ubuntu instance on AWS
- Configure security group (Ports: 22, 80)

- Connect via SSH
- ---
- 2️⃣ **Install Nginx**
- Install and start Nginx
- Verify using public IP
- ---
- 3️⃣ **Install PHP (FPM)**
- Install PHP and PHP-FPM
- Configure Nginx to handle .php files
- ---
- 4️⃣ **Configure Web Directory**
- Setup /var/www/html
- Create index.php file
- Verify PHP execution
- ---
- 5️⃣ **Setup RDS Database**
- Create MySQL DB instance
- Configure public access
- Allow EC2 access via security group
- ---
- 6️⃣ **Connect EC2 to RDS**
- Install MySQL client
- Connect using RDS endpoint
- Create database and tables
- ---
- 7️⃣ **Build Dashboard UI**
- Create PHP-based UI

- Add styling (CSS)
- Display connection status

- **8️⃣ Integrate Backend Logic**
- Connect PHP with MySQL
- Validate DB connection
- Show success/failure status

- **🔄 Workflow**
- User accesses the application via browser
- Request reaches EC2 (Nginx server)
- Nginx forwards request to PHP-FPM
- PHP processes logic and queries database
- Data retrieved from RDS (MySQL)
- Response returned to user as dashboard UI

- **📊 Key Features**
- ✓ Cloud-based deployment
  ✓ High-performance Nginx server
  ✓ Secure database integration (RDS)
  ✓ Dynamic PHP backend
  ✓ Interactive dashboard UI
  ✓ Scalable architecture

Final Summary

- **NexHost** is a complete cloud-hosted LEMP application that demonstrates how to deploy and integrate a web server, backend logic, and database using AWS services.
  The project successfully showcases **real-world web hosting architecture**, combining performance, scalability, and simplicity, making it ideal for learning cloud deployment and full-stack fundamentals.

SnapShot :

```
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affiliates. Other names may be trademarks of their respective
owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE DATABASE nexhostdb;
Query OK, 1 row affected (0.08 sec)

mysql> USE nexhostdb;
Database changed
mysql>
mysql> CREATE TABLE users (
    ->     id INT AUTO_INCREMENT PRIMARY KEY,
    ->     username VARCHAR(50),
    ->     password VARCHAR(50)
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql>
mysql> INSERT INTO users (username, password)
    -> VALUES ('vivek', '1234');
Query OK, 1 row affected (0.01 sec)

mysql> sudo nano /var/www/html/index.php
    -> exit
    ->
    -> ^C
mysql> exit
Bye
ubuntu@ip-172-31-6-100:~$ sudo nano /var/www/html/index.php
ubuntu@ip-172-31-6-100:~$
```

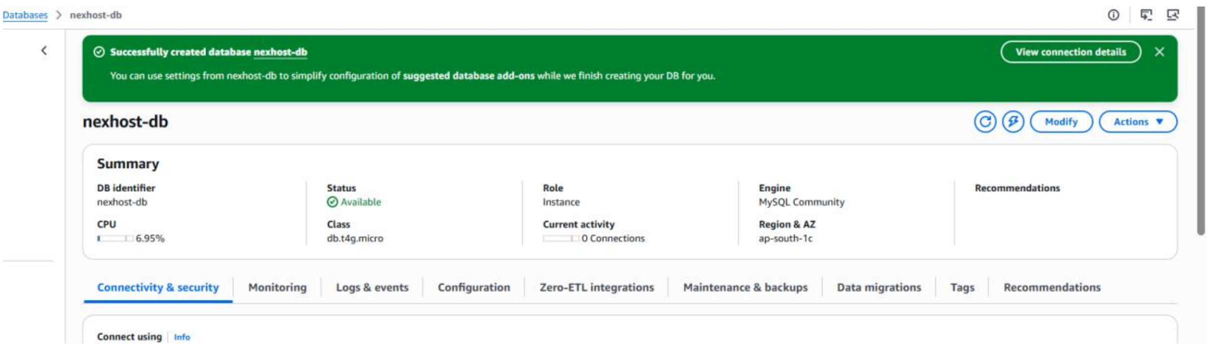

Before RDS :

After RDS : (Final Output)